

COURSE TITLE : ADVANCED COMPUTER APPLICATION LAB
COURSE CODE : 6189
COURSE CATEGORY : A
PERIODS/WEEK : 3
PERIODS/SEMESTER : 45
CREDITS : 2

TIME SCHEDULE

MODULE	TOPIC	PERIODS
I	3D STUDIO MAX.	8
II	MODIFYING AND EDITING COMMANDS IN 3DSMAX	15
III	MAPPING, RENDERING , LIGHTS AND CAMERA	8
IV	MS PROJECT	14
TOTAL		45

Course Objectives:

Sl.	Sub	Student will be able to
1	1	Develop 3D views from 2D drawings
	2	Apply materials to the prepared views
	3	View the drawing in different angles
2	1	Take print outs using printers and plotters in different scales.
	2	Import drawings from ACAD to 3D STUDIO and apply materials , background, lighting.
	3	Locate camera positions and make walk throughs.
3	1	Use MS PROJECT

Course Distribution:

Upon completing the course the student should be able to:

MODULE I-- 3D STUDIO MAX

1.1.0. Familiarize with software interface

1.1.1 Set up units

1.1.2. Use main tool bar

MODULE II -- MODIFYING AND EDITING COMMANDS IN 3DSMAX

2.1.0. Familiarize with modifiers

2.2.0. Know the concept of 2D shapes

2.3.0. Modify solids and 2D shapes

MODULE III -- MAPPING, RENDERING, LIGHTS AND CAMERA

3.1.0. Familiarize mapping and material application

3.2.0. Know importing and exporting 3D files

3.3.0. Render views

3.4.0. Use lights, cameras and walk through

MODULE IV-- MS PROJECT

4.1.0. Familiarize with Scheduling

4.2.0. Familiarize with Task linkage

4.3.0. Know Time estimate

4.4.0. Know Critical path

4.5.0. Familiarize with Resource management

CONTENT DETAILS

MODULE I--INTRODUCTION TO 3DS MAX

Uses In Various Engineering Sectors-Features In The Software Interface-Four Ports And Configuration - Unit Setup,-Application Of Objects Like Standard Primitive, Extended Primitives -Application Of Tools In Main Tool Bar-Link, Unlink, Selection Tools, Operating Tools, Mirror & Snaps -Software Interface- Navigation Panel. Tab Panel, Media Panel, Grid and Snap Set Up-Panel - Create, Modify, Hierarchy, Object Categories. Tools- Array , Mirror, Move, Copy, Rotate, Scale etc-Exercise- Table, chair, hut, simple compound wall etc.

MODULE II -- MODIFYING AND EDITING COMMANDS IN 3DSMAX

Modifier(Bend ,Edit poly & mesh, Taper , Twist , Shell , Slice , Wave , Noise , Extrude, Lathe , Loft, Boolean Etc)-Concepts of 2d Shapes -Spline (Circle , Ellipse , Helix , Text , Stares) And Default Shapes ,Brief Visualization On Arc , Rec , Ngon , -Modifying Parameters of Line -. Vertex; Segments; Spline, - Creating 360° Circular 3d Objects By Lathe Command -Creating Single & Multiple Boolean (Subtraction Of Doors & Windows From Walls)-Creating Single & Multiple Loft- Exercise- Pillar, Door, Window, Flower vase etc.

MODULE III –MAPPING, RENDERING , LIGHTS AND CAMERA

Introduction to Material Textures and Maps Concepts of Texturing And Adding Material-Introduction To Material Editor-Mapping Material Slots -Color Concepts Texturing With Bitmap Files-Apply Material On Surfaces -UVW Mapping Tiling -Bitmap Material Creating Mirror And Glossiness -Multi Material Editing Maps-Relation With Auto Cad And Line Drawing-Importing 2d Plans And 3d Blocks From AutoCAD - Exporting 3d Max Files To Other Extension-Concepts Of Rendering - Scan Line Rendered, Rendering from Different Views-Environments and Back Grounds.-Active View Port Settings-Creating Still Images (Jpeg, Bmp)-, Creating Lights In View Ports-Concepts of Walk Through, camera-exercise – Modeling of buildings.

MODULE IV--M S Project.

Scheduling-Task linkage-Time estimate-Critical path-Resource management